ITA14 – ETHICAL HACKING

ASSIGNMENT 2 – CO2

Assignment Title: Evaluation and Engineering of Footprinting, Reconnaissance and Social-Engineering-Awareness Workflows in an Authorized Assessment

CO2: Execute footprinting and social-engineering techniques such as DNS analysis, competitive intelligence gathering and attack simulations to collect information for identifying potential security risks.

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

Submission Date: _____

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Department: INFORMATION TECHNOLOGY

ASSIGNMENT QUESTION PAPER

Scenario – Assessment Design Philosophy

The assignment is designed for high-performing cybersecurity students and emphasizes the evaluation of reconnaissance and information-gathering choices rather than the mechanical use of tools. Students are expected to formulate an information-gathering strategy, construct difficult test conditions, run footprinting against authorized targets, collect evidence, identify limitations, compare alternatives and defend engineering decisions. All work is performed only against authorized or lab-owned targets, and no attack is launched against live systems or real individuals.

The assessment follows: Understand → Predict → Design → Collect → Analyse → Evaluate → Defend → Validate.

Q1 – Evaluate and Engineer a Footprinting Strategy

Problem Context – Multi-Source Open-Source Information Gathering

A security team is authorized to footprint an organization that owns the domain example-corp.example before a penetration test. Each engagement must gather information from WHOIS, DNS, search engines and public documents, organize it, and use it to understand the target's attack surface.

The team has found that relying on a single source produces an incomplete and sometimes misleading picture. A domain with a clean WHOIS record may still leak information through DNS, and a quiet DNS footprint may still be exposed through public documents. The strategy must therefore combine several sources and judge their reliability rather than trusting one.

Assessment Challenge

- Design a footprinting strategy that combines passive and active techniques (WHOIS, DNS records, search-engine operators and archived pages) and classify each technique as passive or active.
- Collect information from an authorized domain using appropriate tools such as whois, nslookup and dig, and organize the findings by source.
- Ensure that out-of-scope domains, personal data and unverified results are not treated the same as authorized, verified, in-scope findings.
- Construct four contrasting targets: a well-protected domain; a domain that leaks information; a domain with misconfigured DNS; and a domain with a minimal public footprint.
- For every target, predict what information will be found before collection and compare it with the actual result.
- Record the sources queried and the records found, and identify cases where one source contradicts another.
- Evaluate whether the chosen sources are reliable and whether any source produces misleading or outdated information.

Changed Condition: The scope grows from one domain to fifty subdomains and several third-party services. Evaluate the original strategy, modify it where necessary, and defend whether it remains suitable for large-scale authorized reconnaissance.

Evidence Expected: Footprinting strategy design, authorized test targets, collected-source records, prediction-versus-observation analysis, source-reliability analysis, scope analysis and a justified final strategy.

Q2 – Build a Reliable DNS Footprinting Workflow

Problem Context – Conflicting and Misconfigured DNS Data

An authorized assessment must map a target's DNS by retrieving A, MX, NS and TXT records and, only where explicitly permitted, attempting a zone transfer. Records may be inconsistent, delegated to third parties, cached or misconfigured, so the workflow must interpret the results carefully rather than reporting them at face value.

Assessment Challenge

- Develop a DNS footprinting workflow using dig and nslookup that retrieves A, MX, NS and TXT records, and attempts a zone transfer only where authorized.
- Evaluate how split-horizon DNS, delegation and caching change the information visible to an external tester.
- Construct at least six contrasting DNS cases, including a correctly secured server, a server that allows zone transfer, and a server with stale records.

- For each case, predict the records exposed, run the workflow, trace what each query returns, and compare the result with the prediction.
- Evaluate whether a misconfigured zone transfer meaningfully changes the attacker's knowledge and the resulting risk.
- Introduce a countermeasure such as restricting transfers or using TSIG, and redesign the analysis to show the reduced exposure.

Evidence Expected: DNS workflow, record analysis, authorized test cases, prediction-versus-observation, zone-transfer risk evaluation and countermeasure redesign.

Q3 – Create a Competitive-Intelligence Awareness Utility

Problem Context – Public Information and Human-Risk Exposure

A department is building a repeatable workflow that collects public business information about an authorized organization from job advertisements, annual reports, press releases and public profiles, and assesses how that information could support social-engineering attacks such as phishing, vishing, pretexting and tailgating, so that awareness and countermeasures can be designed. No social-engineering attack is performed against real people.

Your Challenge

- Design documented steps for each public source and record where each item came from, using search-engine operators such as site:, filetype: and intitle:.
- Map each collected item to the social-engineering technique it could enable (phishing, vishing, pretexting or tailgating) and the indicator a target should notice.
- Create test organizations containing rich, sparse and misleading public information.
- Predict the information and the human-risk level expected from each source before collection.
- Evaluate whether the collected information meaningfully increases social-engineering risk and which source is the most dangerous.
- Add two new sources or two new awareness countermeasures without restructuring the whole workflow, and evaluate the effect.
- Record the sources queried, the items found and the countermeasures recommended.

Inquiry: Design and implement a modular competitive-intelligence and awareness workflow using documented sources and operators, and evaluate the human attack surface of an authorized organization.

Evidence Expected: Workflow architecture, source operators, item-to-technique mapping, test organizations, prediction-versus-observation, risk evaluation and recommended countermeasures.

Q4 – Design a Reconnaissance Data-Validation Mechanism

Problem Context – High-Stakes Reconnaissance Data Quality

A reconnaissance data-collection process receives results that may be incomplete, outdated, duplicated or out of scope. Acting on poor-quality data wastes effort and can breach the agreed scope, so each item must be validated before it is used.

Your Challenge

- Develop a validation mechanism that checks each collected item for scope, freshness, duplication and source reliability.
- Distinguish a recoverable item (one that needs a re-query or confirmation) from an item that must be rejected (out of scope or unverifiable).
- Construct at least five test sequences, including all-valid, out-of-scope, stale and duplicated data.
- Predict how many items pass, fail or need re-checking before running the validation.
- Evaluate whether the order of the validation checks affects effort and accuracy.
- Redesign the validation order to reduce unnecessary checks while still keeping out-of-scope data out.

Scalability Inquiry: Predict and analyse the validation outcomes and the effort required as the volume of collected data grows from one target to fifty targets.

Evidence Expected: Validation design, adversarial reconnaissance sequences, pass/fail/re-check analysis, ordering evaluation and a justified redesign.

Q5 – Evaluate Footprinting of a Hierarchical Organization

Problem Context – Irregular Multi-Level Organizational Footprint

A target organization has a nested public footprint of a parent domain, subdomains, departments and third-party services. Footprinting must traverse this hierarchy level by level, recording each level and stopping cleanly at out-of-scope or terminal nodes.

Your Challenge

- Represent the organization's public footprint as a hierarchy (domain, subdomains and services) that can be traversed systematically.
- Develop a systematic traversal that records each level and stops at out-of-scope or terminal nodes using an explicit boundary condition.
- Construct shallow, deeply nested and irregular footprints with branches of differing size.
- Predict the number of nodes and the information expected before the traversal.
- Evaluate how depth and branching affect the effort and completeness of the footprinting.
- Include an empty branch or a node with no public data and evaluate whether the boundary condition handles it correctly.
- Compare the systematic traversal conceptually with an ad-hoc, unstructured approach.

Test Your Design: Verify whether the traversal correctly covers shallow, deep and irregular footprints without leaving the authorized scope.

Correctness Inquiry: Predict and verify the expected coverage, the boundary handling and the number of nodes visited.

Evidence Expected: Hierarchical footprint representation, traversal design, test structures, prediction-versus-observation, boundary analysis and a justified final design.

Q6 – Evaluate Footprinting Approaches Under Constraints

Problem Context – Reconnaissance Under Time and Stealth Constraints

An authorized assessment must gather enough information while balancing speed, completeness and the risk of detection. Passive open-source collection and more active footprinting have different trade-offs, so the better approach depends on the scope and the constraints.

Your Challenge

- Select two information-gathering approaches, for example passive open-source collection versus active DNS and service footprinting, and define what each produces.
- Apply both approaches to the same authorized target.
- Test at least four increasing scope sizes, including one at which one approach becomes impractical.
- Record the number of queries or steps and the time taken where useful.
- Evaluate redundant or repeated collection in each approach and identify where effort is wasted.
- Determine whether the faster or the quieter approach is always the better engineering choice.
- Defend a recommendation for a small-scope engagement and for a large-scope engagement.

Inquiry: The workflow is used both for teaching and for repeated real assessments; justify the chosen approach for each context.

Evidence Expected: Approach definitions, two workflows, quantitative measures, redundancy analysis and a defended recommendation.

DAY-4 DESIGN CHALLENGE

Q7 – Integrated Review: Diagnose, Redesign and Defend

Select one workflow from Q1–Q6 that demonstrates a meaningful limitation, such as incomplete coverage, an unreliable source, excessive queries, weak validation or a risk of leaving scope, and improve it using evidence.

A. Diagnose the Limitation

- Identify the reconnaissance or logical bottleneck using evidence.
- Determine whether the limitation is caused by source unreliability, redundant collection, poor ordering or a risk of going out of scope.

Evidence Expected: Identify the reconnaissance or logical bottleneck and support the diagnosis with evidence. Pinpoint the underlying cause of the limitation, such as redundant collection, an unreliable source or poor ordering.

B. Establish Evaluation Criteria

- Define measurable criteria against which the original workflow will be judged.
- Examples include coverage, the number of queries, source reliability, scope compliance and time.

Evidence Expected: A clearly defined set of measurable evaluation criteria, including coverage, number of queries, source reliability and scope compliance.

C. Propose and Implement a Redesign

- Modify the original workflow to address the diagnosed limitation.
- Provide the design logic or the ordered steps before implementation.
- Identify any trade-off introduced by the redesign.

Evidence Expected: A redesigned workflow supported by design logic or ordered steps, demonstrating how the diagnosed limitation is addressed and identifying the trade-offs introduced.

D. Validate and Compare

- Use equivalent test conditions for the original and the redesigned workflow.
- Collect quantitative evidence and determine whether the redesign actually improves the selected criterion.

Evidence Expected: A quantitative comparison of the original and redesigned workflows under equivalent conditions, using measurable criteria to verify the effectiveness of the redesign.

E. Defend the Final Engineering Judgment

- Evaluate whether the redesign met its objective.
- Explain cases in which the original workflow may still be preferable.
- Defend the final choice using experimental evidence rather than subjective statements.

Evidence Expected: A justified final engineering decision supported by experimental results, evaluating the redesign's effectiveness and identifying cases in which the original workflow remains preferable.

MARK DISTRIBUTION

Question	Assessed Area
Q1	Footprinting strategy design and evaluation
Q2	DNS footprinting and analysis workflow
Q3	Competitive-intelligence and awareness utility
Q4	Reconnaissance data validation and recovery
Q5	Hierarchical organization footprinting
Q6	Footprinting approach comparison
Q7	Integrated diagnose–redesign–validate challenge
TOTAL	

Submission Requirements

Submit:

- Assignment report containing evidence for Q1–Q7.
- Scripts or documented command sequences for all implemented workflows.
- Student-designed authorized test targets and adversarial cases.
- Raw collected data and tool outputs.
- Prediction-versus-observation comparisons.
- Tables/graphs wherever they strengthen the evaluation.
- Source, query, item and scope counts where applicable.
- Design representation or pseudocode for redesigned workflows.

Execution time may be reported as supporting evidence, but conclusions should not rely solely on timing; coverage, source reliability and scope compliance should form the primary evidence.

ANSWERS

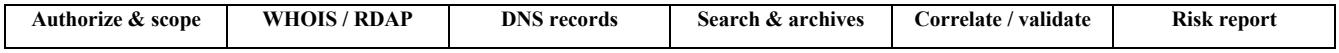
Q1. Evaluate and Engineer a Footprinting Strategy

A footprinting strategy combines passive open-source collection with controlled active queries. Its purpose is to build a verified picture of the authorized organization's public attack surface while avoiding unnecessary contact with systems and excluding out-of-scope information.

1.1 Proposed strategy

Stage	Method / source	Mode	Information collected	Validation
1. Scope	Written authorization and domain list	Planning	Root domain, permitted subdomains and exclusions	Compare every candidate with signed scope
2. Registration	WHOIS / RDAP	Passive	Registrar, registration dates, name servers	Check source date and consistency
3. DNS review	A, MX, NS and TXT queries	Active query	Address, mail routing, name servers and policies	Record resolver, TTL and timestamp
4. Public search	site:, filetype:, intitle:	Passive	Public documents and technology clues	Open and verify original source
5. Archive review	Public web archives	Passive	Historical pages and content	Mark historical data as unconfirmed
6. Correlation	Compare and deduplicate	Analysis	Conflicts, repeated hosts and stale entries	Confirm important findings independently

Figure 1. Multi-source footprinting workflow



1.2 Passive and active techniques

- Passive footprinting includes WHOIS/RDAP lookups, search-engine results, public reports, archived pages and publicly available records. These methods generally do not probe the target host.
- Active footprinting includes direct DNS queries and explicitly approved service checks. Such traffic may be logged by the target or provider.
- Classification depends on how information is obtained. A DNS query to a public resolver is still an active query.

1.3 Four contrasting test targets

Target	Prediction	Illustrative observation	Interpretation
A – Well protected	Limited public records	Privacy-protected registration; expected DNS records	Low visible exposure; not proof of no vulnerabilities
B – Information leakage	Public files may expose roles or technology	A public PDF contains department names and old technology references	Sanitize documents and metadata
C – DNS misconfiguration	Unexpected or excessive DNS data	An authorized lab zone returns an unintended test host record	Verify ownership and impact
D – Minimal footprint	Few indexed pages and basic DNS	Only website and mail records observed	Low yield; greater uncertainty

These are illustrative lab outcomes, not claims about a real organization. Replace them with dated command outputs from the approved test domain.

1.4 Source reliability and scaling

Source	Reliability	Limitation	Best practice
WHOIS / RDAP	High for registration metadata	Privacy masking and update delays	Record lookup time; confirm DNS separately
DNS	High for the queried resolver at that moment	Caching, split-horizon and delegation	Repeat approved queries and note TTL
Search engines	Medium	Indexing delay and ranking bias	Verify the original document
Web archives	High for historical snapshot	Not proof of current exposure	Label with snapshot date

For fifty subdomains, maintain an asset inventory, batch queries with rate limits, deduplicate hostnames and group them by parent domain or provider. Third-party services are included only when authorization explicitly names them. A scalable workflow logs query counts, errors, timestamps and scope decisions.

Engineering judgment: use multiple sources rather than relying on WHOIS or DNS alone. Passive collection should lead; active queries should be targeted, rate-limited and authorized.

Q2. Build a Reliable DNS Footprinting Workflow

DNS footprinting identifies how a domain maps to network services. Retrieve A, MX, NS and TXT records, compare the results and attempt a zone transfer only when written permission explicitly allows it.

Purpose	Example command	Information revealed
IPv4 address	dig example-corp.example A	Address returned for the name
Mail servers	dig example-corp.example MX	Mail exchangers and priorities
Authoritative DNS	dig example-corp.example NS	Name-server delegation
Text records	dig example-corp.example TXT	SPF and other published text
Alternative query	nslookup -type=MX example-corp.example	Mail-server result using another utility
Zone transfer – lab only	dig @ns1.example-corp.example example-corp.example AXFR	Whether the lab DNS server permits transfer

Figure 2. DNS collection workflow

Confirm scope	Query A / MX	Query NS / TXT	Check TTL & authority	Compare answers	Document

2.1 Six contrasting DNS cases

Case	Expected result	Interpretation / action
Secure server	Normal records; AXFR refused	Record refusal as expected configuration
AXFR allowed in lab	Zone records returned	Restrict transfer to approved secondary servers; use TSIG
Stale A record	Old address in one resolver cache	Compare TTL and authoritative answer
Split-horizon DNS	Internal and external answers differ	Record vantage point; difference may be intentional
Third-party delegation	NS points to managed DNS provider	Confirm provider is in scope
Incomplete TXT policy	SPF or verification text absent or outdated	Report observed facts; recommend owner review

2.2 DNS behaviour and countermeasures

- Split-horizon DNS intentionally supplies different answers to internal and external clients.
- Delegation assigns authority for a child zone to other name servers; follow only authorized zones.
- Caching stores answers for the TTL period, so a resolver may temporarily return an older value.
- An unintended zone transfer can expose hostnames and infrastructure relationships, but does not itself prove a service is vulnerable.

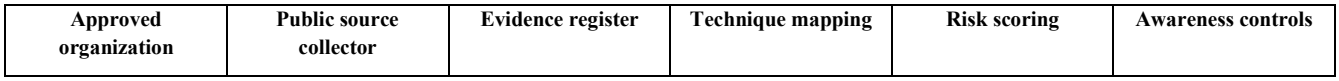
Control	Purpose
Allow AXFR only to named secondary servers	Prevents arbitrary clients from downloading the zone
Use TSIG where supported	Authenticates transfers between trusted servers
Review public DNS records	Removes obsolete or accidental records
Monitor DNS logs	Detects unexpected transfer attempts
Separate internal and public zones	Reduces disclosure of internal naming

Conclusion: record resolver, timestamp, record type, TTL, authoritative source and authorization boundary.

Q3. Create a Competitive-Intelligence Awareness Utility

Competitive intelligence collects information intentionally made public. The aim is to understand how public information could be misused in social engineering and then design awareness measures. No message, call or deception is directed at a real person.

Figure 3. Awareness utility architecture



3.1 Public sources and search operators

Source	Search operator	Possible information	Safe handling
Job advertisements	site:careers.example filetype:pdf	Roles, tools and team structure	Use published job content
Annual reports	site:example.org filetype:pdf	Departments and leadership roles	Record report year and page
Press releases	site:example.org intitle:press release	Projects, vendors and events	Verify date and publisher
Public profiles	site:example.org team	Public job titles and business contacts	Avoid personal or sensitive details

3.2 Information-to-technique mapping

Public item	Technique	Warning indicator	Countermeasure
Project announcement	Pretexting	Caller claims to be a partner and asks for access	Verify through known channels
Published email format	Phishing	Unexpected email imitates a familiar address	Check sender and report suspicious links
Public office number	Vishing	Caller requests credentials or OTP urgently	Never disclose passwords or OTPs
Visitor information	Tailgating	Unknown person follows staff through a secure entrance	Follow visitor policy

3.3 Test organizations and awareness measures

Organization	Public profile	Predicted risk	Reason
A – Rich	Reports, job posts and project announcements	Higher	Many current clues may support impersonation
B – Sparse	Basic website and few reports	Lower to moderate	Fewer clues, but basic impersonation remains possible
C – Misleading	Old pages and inaccurate archived content	Variable	Outdated facts can create false leads

- Add a vendor-directory verification step: confirm supplier requests using a phone number already stored in the official directory.
- Review document metadata before publication and remove hidden author names, comments and unnecessary internal properties.
- Run awareness simulations only with institutional approval, fictional identities and a controlled training platform. Never collect real passwords or OTPs.

The most dangerous source is not necessarily the one with the most records. A small amount of current information about a project or supplier may be more useful than many outdated pages.

Q4. Design a Reconnaissance Data-Validation Mechanism

Validate each collected item for authorization scope, freshness, duplication and source reliability. Assign one of three outcomes: PASS, RE-CHECK or REJECT.

Figure 4. Ordered validation mechanism

Receive item	Scope check	Duplicate check	Freshness check	Source check	Final decision

Check	Pass	Re-check	Reject
Scope	Within approved domain	Ownership unclear	Clearly out of scope
Duplicate	Unique normalized item	Possible alias	Exact duplicate
Freshness	Within defined age limit	Date missing or near expiry	Known obsolete and irrelevant
Reliability	Primary or independently confirmed	Weak or conflicting source	Unverifiable or fabricated

4.1 Test sequences and predicted outcomes

Sequence	Input	PASS	RE-CHECK	REJECT
All valid	5 valid unique records	5	0	0
Out of scope	3 valid + 2 external names	3	0	2
Stale data	3 current + 2 old records	3	2	0
Duplicates	4 unique + 2 repeated entries	4	0	2
Mixed	2 valid, 1 stale, 1 duplicate, 1 out of scope	2	1	2

These are designed test inputs and predicted outcomes. Actual results should be recorded by executing the validator against the same test set.

4.2 Pseudocode

FOR each item in collected_data:\n IF item is outside authorized scope: REJECT\n ELSE IF item is an exact duplicate: REJECT\n ELSE IF item date is missing or stale: RE-CHECK\n ELSE IF source is weak or conflicting: RE-CHECK\n ELSE: PASS\n Record decision, reason, source and timestamp

4.3 Efficient ordering and scaling

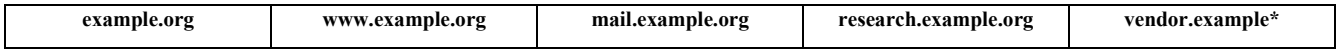
Check scope first because out-of-scope items must not proceed. Exact deduplication is inexpensive and comes next. Freshness and reliability checks follow for remaining items. For fifty targets, automate the scope allowlist, normalization and logging while retaining manual review for uncertain records.

Conclusion: uncertain items are held for confirmation; clearly out-of-scope or unverifiable items are rejected.

Q5. Evaluate Footprinting of a Hierarchical Organization

Represent the approved organization as a tree. The root is the authorized parent domain; child nodes are approved subdomains, departments or explicitly included services. Traversal stops at a boundary where ownership or authorization is absent.

Figure 5. Example authorized footprint hierarchy



*A vendor node is visited only if written scope explicitly includes that third-party service.

5.1 Traversal algorithm

VISIT(node):\n IF node is outside approved scope: STOP this branch\n Record node and evidence\n IF node is terminal OR has no public child data: RETURN\n FOR each approved child:\n VISIT(child)

5.2 Contrasting test structures

Structure	Example	Expected behaviour
Shallow	Root → 3 subdomains	Visit root and approved children; stop at leaves
Deep	Root → A → B → C → D	Follow each level while scope remains valid
Irregular	Root branches have 1, 3 and 0 children	Visit available children; empty branch ends cleanly

Test tree	Predicted nodes	Illustrative visited	Boundary result
Shallow: root + 3 leaves	4	4	All leaves terminal
Deep: root + 4 nested nodes	5	5	Stops at deepest node
Irregular: root + branches + empty branch	4	4	Empty branch returns without error
Mixed: root + 2 in-scope + 1 vendor	4 candidates	3	Vendor stopped as out of scope

5.3 Complexity and comparison

For a tree with N approved nodes, traversal visits each node at most once, giving O(N) time. A depth-first traversal uses O(H) stack memory, where H is maximum depth.

- Systematic traversal provides repeatable coverage, clear evidence and explicit boundary handling.
- Ad-hoc traversal may miss branches, revisit hosts or cross into third parties.
- An empty branch is a valid terminal condition and should be recorded as having no public child data.

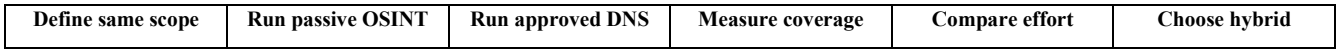
Engineering judgment: use inventory-driven depth-first traversal with a scope check before every child query.

Q6. Evaluate Footprinting Approaches Under Constraints

Passive open-source collection and active DNS/service footprinting have different trade-offs. Passive collection generally creates less direct interaction, while active queries can provide current technical answers but may be logged and require tighter control.

Dimension	Passive OSINT	Active DNS / service footprinting
Inputs	Search engines, reports, archives, public registration	Approved names, DNS resolvers and permitted services
Outputs	Public documents and historical clues	Current DNS answers and approved service metadata
Target interaction	Usually indirect	Direct queries
Strength	Low impact and broad context	Current technical observations
Limitation	Stale or incomplete indexing	Detection, rate limits and scope risk

Figure 6. Comparison workflow



6.1 Controlled comparison plan

Apply both approaches to the same lab-owned domain and approved names. Record query count, elapsed time, unique verified findings, stale findings and out-of-scope rejections. Timing is supporting evidence; completeness and scope compliance are primary.

Scope	Passive steps*	Active queries*	Evaluation focus
1 domain	5	4	Baseline coverage
5 names	12	20	Repeated collection
20 names	35	80	Batching and deduplication
50 names	80	200	Manual effort and rate limits

*Planning estimates only, not measured results. Replace with actual logs after execution.

6.2 Redundancy and recommendations

- Passive search may return the same document through several queries; normalize URLs and retain one canonical record.
- Active DNS may repeat identical names and record types; cache results where permitted and record TTL.
- A fast approach is not always better if it produces stale results or violates scope.

Engagement	Recommended approach	Reason
Small scope	Passive first, then a few approved DNS queries	Context plus technical confirmation at low effort
Large scope	Automated, rate-limited DNS inventory plus passive enrichment	Repeatable coverage and efficient batch processing
Teaching exercise	Passive discovery followed by instructor-controlled lab queries	Students can compare methods safely

Final judgment: use a hybrid strategy and choose based on scope, authorization, freshness and monitoring constraints—not speed alone.

Q7. Integrated Review: Diagnose, Redesign and Defend

Selected workflow: Q4 reconnaissance data validation. Poor ordering can waste effort if analysts verify freshness and source quality for an item that should first have been rejected because it is outside the approved scope.

A. Diagnose the Limitation

The original sequence does not define an efficient order for scope, freshness, duplication and reliability checks. Expensive verification may be performed on irrelevant or repeated records.

B. Establish Evaluation Criteria

Criterion	Measurement
Scope compliance	Out-of-scope items passing the scope gate; target = 0
Efficiency	Validation checks per accepted item
Accuracy	Correct PASS / RE-CHECK / REJECT classification
Coverage	Percentage of in-scope unique items receiving a decision

C. Propose and Implement a Redesign

Figure 7. Redesigned validation sequence

Normalize	Scope allowlist	Deduplicate	Freshness	Source reliability	Log decision

Normalize hostnames and URLs; reject items failing the scope allowlist; remove exact duplicates; then check freshness and reliability. Uncertain ownership, missing dates or conflicting sources lead to RE-CHECK. Store a reason code for every decision.

D. Validate and Compare

Test set	Original checks*	Redesigned checks*	Expected outcome
10 records, 2 out of scope	40	32	Both reject 2; redesign avoids later checks
10 records, 3 duplicates	40	31	Both retain 7 unique records
10 records, 2 stale	40	40	Both send 2 to re-check
10 mixed records	40	30–34	Same classifications with fewer unnecessary checks

*Illustrative estimates, not observed measurements. Execute both algorithms on an identical labelled dataset and replace estimates with measured counts.

E. Defend the Final Engineering Judgment

The redesign is preferable for large collections because it enforces scope first and removes duplicates before expensive checks. The original sequence may remain adequate for a small manually reviewed dataset where all items are already confirmed in scope and duplicates are rare. Choose using measured coverage, accuracy, check count and zero scope violations.

Overall Conclusion

Footprinting is not simply collecting the largest number of records. A sound authorized assessment combines multiple sources, interprets DNS behaviour, understands human risk from public information, validates every item, traverses organizational hierarchies within strict boundaries and selects an approach suited to the engagement. The strongest design is repeatable, evidence-based, proportionate and auditable.

References and Practical Command Notes

- WHOIS / RDAP: use the relevant registry or registrar lookup service for public registration information.
- DNS utilities: dig and nslookup for A, MX, NS and TXT record queries on authorized domains.
- Search operators: site:, filetype: and intitle: to narrow public search results.
- Use only lab-owned or explicitly authorized targets. Do not attempt zone transfers or social-engineering interactions against real third parties without written authorization.

- All example targets, case outcomes and quantitative counts are illustrative; insert actual lab evidence where required by the instructor.

Assessment Verification / Viva

Students may be asked to explain and defend their source selection, footprinting methodology, DNS and open-source analysis, scope and authorization decisions, social-engineering-awareness reasoning, adversarial test design, measured results, redesign choices and final engineering judgments.

Overall Assessment Expectation

The assignment assesses the student's ability to evaluate information-gathering solutions under realistic, authorized conditions: Problem Interpretation → Prediction → Design → Collection → Adversarial Testing → Measurement → Analysis → Evaluation → Defence → Engineering Judgment.